Finiteness of the Extension of $\mathbb{Q}$ Generated by Frobenius Traces, in Finite Characteristic

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Dedicated to the memory of I. M. Gelfand

Abstract. Let Z_0 be a scheme of finite type over $\mathbb{F}_q$ and let $\mathcal{F}_0$ be a $\overline{\mathbb{Q}}_\ell$-sheaf on Z_0. We show that there is a finite-type extension $E \subset \overline{\mathbb{Q}}_\ell$ of $\mathbb{Q}$ such that the local factors of the L-function of $\mathcal{F}_0$ all have coefficients in E. If Z_0 is normal and connected and $\mathcal{F}_0$ is an irreducible ℓ-adic local system whose determinant has finite order, one may take E to be a finite extension of $\mathbb{Q}$.

Contents.

0. Introduction

0.0

Fix the following notation:

$$\begin{aligned}
\mathbb{F}_q &: \text{a finite field with } q \text{ elements, of characteristic } p;\\
\mathbb{F} &: \text{an algebraic closure of } \mathbb{F}_q;\\
\mathbb{F}_{q^n} &: \text{the extension of degree } n \text{ of } \mathbb{F}_q \text{ inside } \mathbb{F};\\
W(\mathbb{F}/\mathbb{F}_{q^n}) &: \text{the subgroup } \mathbb{Z} \text{ of } \mathrm{Gal}(\mathbb{F}/\mathbb{F}_{q^n}) = \widehat{\mathbb{Z}} \text{ generated by the geometric Frobenius } F\\
&\qquad \text{(the inverse of } x \mapsto x^{q^n});\\
\ell &: \text{a prime number distinct from } p;\\
\overline{\mathbb{Q}}_\ell &: \text{an algebraic closure of } \mathbb{Q}_\ell.
\end{aligned}$$

An object over $\mathbb{F}_q$ will be marked by a subscript 0, and suppressing this subscript will mean extension of the base field from $\mathbb{F}_q$ to $\mathbb{F}$.

For a summary of the formalism of $\overline{\mathbb{Q}}_\ell$-sheaves and of $\overline{\mathbb{Q}}_\ell$-Weil sheaves, I refer to [De2] 1.1.1 and 1.1.10.

Let $\overline{X}_0$ be a smooth projective absolutely irreducible curve over $\mathbb{F}_q$, let S_0 be a finite set of closed points of $\overline{X}_0$, and put $X_0 := \overline{X}_0 - S_0$. We write K_0 for the field of rational functions on X_0 and A for its adele ring.

0.1

Lafforgue [L] provides a natural bijection between

(A) isomorphism classes of smooth irreducible $\overline{\mathbb{Q}}_\ell$-Weil sheaves of rank r on X_0, and

(B) cuspidal automorphic representations π of $\mathrm{GL}(r, A)$, unramified outside S_0.

This bijection is characterized by equality, at each closed point x_0 of X_0, of the local factors of the L-functions attached to the objects in (A) and (B).

In (B), the notion of cuspidal automorphic representation is purely algebraic: one may take as coefficient field any algebraically closed field k of characteristic zero. We take $k = \overline{\mathbb{Q}}_\ell$. The topology of $\overline{\mathbb{Q}}_\ell$ plays no role.

If Z_0 is a scheme of finite type over $\mathbb{F}_q$, the action on $\mathbb{F}$ of $W(\mathbb{F}/\mathbb{F}_q)$ induces an action of this Weil group on $Z = Z_0 \otimes_{\mathbb{F}_q} \mathbb{F}$. A $\overline{\mathbb{Q}}_\ell$-Weil sheaf $\mathcal{F}_0$ on Z_0 is a $\overline{\mathbb{Q}}_\ell$-sheaf $\mathcal{F}$ on Z, endowed with an action of $W(\mathbb{F}/\mathbb{F}_q)$ on $(Z, \mathcal{F})$ lifting its action on Z. The notion of $\overline{\mathbb{Q}}_\ell$-sheaf is defined by passage to the limit from that of an etale sheaf of $\mathcal{O}_\lambda/I$-modules, where $\mathcal{O}_\lambda$ is the integral closure of $\mathbb{Z}_\ell$ in a finite extension $E_\lambda \subset \overline{\mathbb{Q}}_\ell$ of $\mathbb{Q}_\ell$, and I is a power of the maximal ideal. Here the topology of $\overline{\mathbb{Q}}_\ell$ plays a crucial role. A surprising consequence of the equivalence between (A) and (B) is that, for semisimple smooth sheaves on curves, this role is illusory.

0.2

One should beware that Lafforgue assumes a choice of an isomorphism $\iota : \mathbb{C} \to \overline{\mathbb{Q}}_\ell$, and uses it rather to compare automorphic representations in the classical sense, with coefficient field $\mathbb{C}$, and $\overline{\mathbb{Q}}_\ell$-sheaves. If one uses ι to replace cuspidal automorphic representations with coefficients in $\mathbb{C}$ by those with values in $\overline{\mathbb{Q}}_\ell$, the resulting correspondence depends, when r is even and q is an odd power of p, on the square root $\iota(\sqrt{q})$ of q in $\overline{\mathbb{Q}}_\ell$. Changing $\sqrt{q}$ to $-\sqrt{q}$ changes the correspondence by composing it with twisting, see 0.3, by the character

$$n \longmapsto (-1)^{n(r-1)}$$

of $W(\mathbb{F}/\mathbb{F}_q) = \mathbb{Z}$.

Instead of the correspondence used by Lafforgue, we prefer its twist by the character

$$n \longmapsto (\sqrt{q})^{n(r-1)}$$

of $W(\mathbb{F}/\mathbb{F}_q)$. This rationalized correspondence gives a bijection between (A) and (B) independent of ι.

0.3

For both the objects (A) and the objects (B), twisting by a character

$$\chi : W(\mathbb{F}/\mathbb{F}_q) = \mathbb{Z} \longrightarrow \overline{\mathbb{Q}}_\ell^*$$

is defined, and these twists correspond.

The set of characters χ of $W(\mathbb{F}/\mathbb{F}_q)$ is naturally in bijection with the set of isomorphism classes of rank-one $\overline{\mathbb{Q}}_\ell$-Weil sheaves $\mathcal{L}_0$ on $\mathrm{Spec}(\mathbb{F}_q)$; for the objects (A), the twist is given by tensor product with the inverse image, under $X_0 \to \mathrm{Spec}(\mathbb{F}_q)$, of such a sheaf.

Let $v : A^* \to \mathbb{Z}$ be the homomorphism such that $\|x\| = q^{-v(x)}$. It factors through the idele class group $K_0^* \backslash A^*$. In (B), if π is a cuspidal automorphic representation of $\mathrm{GL}(r, A)$, viewed as contained in the space of locally constant functions on $\mathrm{GL}(r, K_0) \backslash \mathrm{GL}(r, A)$, the twist π_χ of π by χ is the space of functions

$$f(x)\, \chi(v(\det x))$$

for f in π.

We will sometimes say "twist by b" for "twist by the character χ of $\mathbb{Z}$ such that $\chi(1) = b$."

0.4

The $\overline{\mathbb{Q}}_\ell$-sheaves form a full abelian subcategory of the abelian category of $\overline{\mathbb{Q}}_\ell$-Weil sheaves. On $\mathrm{Spec}(\mathbb{F}_q)$, the isomorphism class of a rank-one $\overline{\mathbb{Q}}_\ell$-Weil sheaf $\mathcal{L}_0$ is determined by a character

$$\chi : W(\mathbb{F}/\mathbb{F}_q) = \mathbb{Z} \longrightarrow \overline{\mathbb{Q}}_\ell^*,$$

and for $\mathcal{L}_0$ to be a $\overline{\mathbb{Q}}_\ell$-sheaf it is necessary and sufficient that χ extend to a continuous character of $\mathrm{Gal}(\mathbb{F}/\mathbb{F}_q) = \widehat{\mathbb{Z}}$, i.e. that $\chi(1)$ be an ℓ-adic unit. This is what forces the use, in 0.1(A), of $\overline{\mathbb{Q}}_\ell$-Weil sheaves. If one wanted in (A) to restrict oneself to $\overline{\mathbb{Q}}_\ell$-sheaves, then in (B) one would have to restrict oneself to cuspidal automorphic representations π whose central character ω_π extends to a continuous character of the profinite completion of the idele class group. This would destroy the purely algebraic character of (B).

The following theorems allow one to pass from $\overline{\mathbb{Q}}_\ell$-sheaves to $\overline{\mathbb{Q}}_\ell$-Weil sheaves.

(a) Let Z_0 be a connected normal variety of finite type over $\mathbb{F}_q$, and let $\mathcal{F}_0$ be a smooth irreducible $\overline{\mathbb{Q}}_\ell$-Weil sheaf of rank r on Z_0. Then $\mathcal{F}_0$ is a $\overline{\mathbb{Q}}_\ell$-sheaf if and only if

$$\det(\mathcal{F}_0) := \bigwedge^r \mathcal{F}_0$$

is one ([De2] 1.3.14).

(b) Let $\mathcal{L}_0$ be a smooth rank-one $\overline{\mathbb{Q}}_\ell$-Weil sheaf on Z_0. Definition: $\mathcal{L}_0$ is of finite order if there is an $N > 0$ such that $\mathcal{L}_0^{\otimes N}$ is isomorphic to the constant sheaf $\overline{\mathbb{Q}}_\ell$. There is always an $N > 0$ such that $\mathcal{L}_0^{\otimes N}$ is the inverse image of a Weil sheaf on $\mathrm{Spec}(\mathbb{F}_q)$ ([De2] 1.3.4(i)). The group, for $\otimes$, of isomorphism classes of rank-one Weil sheaves on $\mathrm{Spec}(\mathbb{F}_q)$ is divisible. Every $\mathcal{L}_0$ is therefore the tensor product of an $\mathcal{L}_0'$, of finite order, and the inverse image of a Weil sheaf $\mathcal{L}_0''$ on $\mathrm{Spec}(\mathbb{F}_q)$. For $\mathcal{L}_0$ to be a $\overline{\mathbb{Q}}_\ell$-sheaf it is necessary and sufficient that $\mathcal{L}_0''$ be one.

(c) Every $\mathcal{F}_0$ as in (a) is therefore obtained by twisting from a $\overline{\mathbb{Q}}_\ell$-sheaf $\mathcal{F}_0'$ such that $\det(\mathcal{F}_0')$ has finite order; and $\mathcal{F}_0$ is a $\overline{\mathbb{Q}}_\ell$-sheaf if and only if the twist is tensor product with the inverse image of a rank-one $\overline{\mathbb{Q}}_\ell$-sheaf on $\mathrm{Spec}(\mathbb{F}_q)$.

In [L], Lafforgue states the correspondence between (A) and (B) by restricting in (A) to smooth irreducible $\overline{\mathbb{Q}}_\ell$-sheaves whose determinant has finite order, and by imposing in (B) the corresponding restriction on automorphic representations. The preceding discussion shows that nothing is lost thereby.

0.5. Notation

Let $\mathcal{F}_0$ be a $\overline{\mathbb{Q}}_\ell$-Weil sheaf on Z_0 of finite type over $\mathbb{F}_q$, let x_0 be a closed point of Z_0, and let

$$\deg(x_0) := [k(x_0) : \mathbb{F}_q]$$

be its degree. The choice of $\bar{x}$ in Z above x_0 defines an embedding of $k(x_0)$ into $k(\bar{x}) = \mathbb{F}$. The Weil group $W(\mathbb{F}/k(x_0))$ is generated by the geometric Frobenius F_{x_0}. It is of index $\deg(x_0)$ in $W(\mathbb{F}/\mathbb{F}_q)$, fixes $\bar{x}$, and acts on the fiber $\mathcal{F}_{\bar{x}}$ of $\mathcal{F}_0$ at the geometric point $\bar{x}$. In particular F_{x_0} acts on $\mathcal{F}_{\bar{x}}$. The local factor at x_0 of the L-function of $\mathcal{F}_0$ is

$$\det\left(1 - F_{x_0} t^{\deg(x_0)}, \mathcal{F}_{\bar{x}}\right)^{-1}. \tag{0.5.1}$$

It does not depend on $\bar{x}$, only on x_0.

Let $n \geq 1$ and let x be an $\mathbb{F}_{q^n}$-point of Z_0: $x \in Z_0(\mathbb{F}_{q^n})$. The point x is an $\mathbb{F}_q$-morphism

$$\mathrm{Spec}(\mathbb{F}_{q^n}) \longrightarrow Z_0.$$

Let x_0 be the closed point of Z_0 which is its image, and let $\bar{x}$ be the geometric point

$$\mathrm{Spec}(\mathbb{F}) \longrightarrow \mathrm{Spec}(\mathbb{F}_{q^n}) \xrightarrow{x} Z_0.$$

We write F_x for the "geometric Frobenius" generator of $W(\mathbb{F}/\mathbb{F}_{q^n})$. We have $W(\mathbb{F}/\mathbb{F}_{q^n}) \subset W(\mathbb{F}/k(x_0))$, and

$$F_x = F_{x_0}^{n/\deg(x_0)}. \tag{0.5.2}$$

We will say "the action of F_x (respectively F_{x_0}) on $\mathcal{F}_0$" for "the action of F_x (respectively F_{x_0}) on $\mathcal{F}_{\bar{x}}$." With this abuse of language, the local factor (0.5.1) is written

$$\det\left(1 - F_{x_0} t^{\deg(x_0)}, \mathcal{F}_0\right)^{-1}.$$

It will be convenient for us to consider, rather than these local factors, the traces $\mathrm{Tr}(F_x, \mathcal{F}_0)$ for x in some $Z_0(\mathbb{F}_{q^n})$. The identity of formal series

$$\log\det(1 - ft, V) = -\sum_{n\geq 1} \mathrm{Tr}(f^n, V)\frac{t^n}{n},$$

for an endomorphism f of V, shows that these traces for x above x_0 and the local factor at x_0 determine one another.

Lafforgue [L] VII 6 proves the following theorem, in which (ii) and (iii) result from the equivalence between (A) and (B) of 0.1.

Theorem 0.6 (Lafforgue). Let $X_0/\mathbb{F}_q$ be as in 0.0, and let $\mathcal{F}_0$ be a smooth irreducible $\overline{\mathbb{Q}}_\ell$-Weil sheaf of rank r on X_0, whose determinant $\det(\mathcal{F}_0)$ has finite order.

(i) At every closed point x_0 of X_0, the eigenvalues of F_{x_0} acting on $\mathcal{F}_0$ are Weil numbers of weight 0, i.e. are algebraic numbers of absolute value one at every place not dividing p. In particular $\mathcal{F}_0$ is pure of weight 0.

(ii) There exists a finite extension $E \subset \overline{\mathbb{Q}}_\ell$ of $\mathbb{Q}$ such that, for every closed point x_0 of X_0, the polynomial

$$\det(1 - F_{x_0} t, \mathcal{F}_0)$$

has coefficients in E.

(iii) For E as in (ii), for every prime number $\ell' \neq p$ and every embedding σ of E into $\overline{\mathbb{Q}}_{\ell'}$, there exists a smooth $\overline{\mathbb{Q}}_{\ell'}$-Weil sheaf $\mathcal{F}'_0$ on X_0 such that, at every closed point x_0 of X_0,

$$\det(1 - F_{x_0} t, \mathcal{F}'_0) = \sigma\det(1 - F_{x_0} t, \mathcal{F}_0).$$

The minimal extension E is the field of definition of the automorphic representation π corresponding to $\mathcal{F}_0$, viewed either as an isomorphism class of representations of $\mathrm{GL}(r, A)$ or as a subspace of the space of functions on $\mathrm{GL}(r, K_0)\backslash \mathrm{GL}(r, A)$. Here one must use the rationalized correspondence of 0.2.

0.7

Let Z_0 be a connected normal scheme of finite type over $\mathbb{F}_q$ and let $\mathcal{F}_0$ be a smooth irreducible $\overline{\mathbb{Q}}_\ell$-Weil sheaf on Z_0, whose determinant has finite order. Lafforgue [L] VII 7 shows, by reduction to the case of curves through an argument with plane sections, that assertion (i) of 0.6 remains true for $\mathcal{F}_0$. Beware that the reference to Hartshorne which Lafforgue gives for irreducibility of the restriction of $\mathcal{F}_0$ to a curve X_0 which is a plane section of Z_0 is not adequate: Hartshorne assumes Z_0 projective. We will explain in 1.5–1.9 how to correct the argument, replacing the reference to Hartshorne by a reference to Jouanolou [J].

0.8

Let $\mathcal{F}_0$ and Z_0 be as in 0.7. Using a method developed by Wiesend [W] and Lafforgue's theorem, Drinfeld [Dr] has shown, under the additional hypothesis that Z_0 is smooth, that if $\mathcal{F}_0$ satisfies assertion (ii) of the theorem then $\mathcal{F}_0$ also satisfies (iii). Our aim is to show that every $\mathcal{F}_0$ as in 0.7 satisfies (ii).

0.9

Let us review the sections of the present article. In Section 1, we consider smooth semisimple $\overline{\mathbb{Q}}_\ell$-Weil sheaves on Z_0 normal absolutely irreducible of finite type over $\mathbb{F}_q$. We show how to describe them in terms of smooth $\overline{\mathbb{Q}}_\ell$-sheaves on schemes $Z_0 \otimes_{\mathbb{F}_q} \mathbb{F}_{q^a}$, with determinant of finite order, and whose inverse image on $Z = Z_0 \otimes_{\mathbb{F}_q} \mathbb{F}$ is irreducible. The decomposition (1.4.1) obtained translates (0.3), together with well-known results on the relation between irreducible representations of a group G and of an extension of $\mathbb{Z}$ by G.

In Section 2, we consider a smooth $\overline{\mathbb{Q}}_\ell$-Weil sheaf $\mathcal{F}_0$ on a smooth absolutely irreducible curve X_0 over $\mathbb{F}_q$. For simplicity, we assume X_0 affine. The "complexity" of X_0 will be measured by the integer

$$b_1(X) := \dim H^1_c(X, \overline{\mathbb{Q}}_\ell).$$

We assume $\mathcal{F}_0$ algebraic, i.e. that the traces $\mathrm{Tr}(F_x, \mathcal{F}_0)$, for x in some $X_0(\mathbb{F}_{q^n})$, are algebraic numbers. By 0.6(ii), applied to twists of irreducible subquotients of $\mathcal{F}_0$, we know that these traces generate a finite extension E of $\mathbb{Q}$ in $\overline{\mathbb{Q}}_\ell$. Our aim is to make explicit an integer N, depending on a "complexity" of $(X, \mathcal{F})$, such that E is generated by the $\mathrm{Tr}(F_x, \mathcal{F}_0)$ for x in $X_0(\mathbb{F}_{q^n})$ with $n \leq N$. Put $\log_q^+(a) = \sup(0, \log_q(a))$. If $\mathcal{F}$ is tamely ramified, one obtains N of the form

$$O\big(\log_q^+(b_1(X))\big) + O(1), \tag{0.9.1}$$

where the implicit constants in the O depend on the rank of $\mathcal{F}_0$. For arbitrary $\mathcal{F}_0$, one must replace $b_1(X)$ by

$$b_1(X) + \sum_{s \in S} \alpha_s(\mathcal{F}),$$

where $\alpha_s(\mathcal{F})$ measures the wildness of the ramification of $\mathcal{F}$ at s.

Another way of measuring the wildness of $\mathcal{F}$, much less economical but convenient in Section 3, is in terms of a connected etale cover X' of X on which the inverse image of $\mathcal{F}$ is tamely ramified. We will obtain another N, of the form

$$O\big(\log_q^+(b_1(X'))\big) + O(1). \tag{0.9.2}$$

Here again, the implicit constants in the O depend on the rank of $\mathcal{F}_0$.

The proofs make essential use of 0.6(iii), in order to reduce the posed problem to the following one. For $\mathcal{F}_0$ and $\mathcal{G}_0$ smooth semisimple Weil sheaves on X_0, find N such that if

$$\mathrm{Tr}(F_x, \mathcal{F}_0) = \mathrm{Tr}(F_x, \mathcal{G}_0)$$

for all x in $X(\mathbb{F}_{q^n})$ with $n \leq N$, then $\mathcal{F}_0$ and $\mathcal{G}_0$ are isomorphic.

0.10

The main theorem is proved in Section 3. After a few devissages, one is reduced to the case in which $\mathcal{F}_0$ is a smooth $\overline{\mathbb{Q}}_\ell$-sheaf on a scheme Z_0 which is smooth and absolutely irreducible over $\mathbb{F}_q$, and in which $\mathcal{F}_0$ is algebraic, meaning algebraicity of Frobenius traces. The point is to show that the extension of $\mathbb{Q}$ generated by Frobenius traces is a finite extension. This is done by showing that there exists an integer N such that, for every integer $n > N$ and every point

$x \in Z_0(\mathbb{F}_{q^n})$, the trace $\mathrm{Tr}(F_x, \mathcal{F}_0)$ is contained in the extension of $\mathbb{Q}$ generated by the $\mathrm{Tr}(F_{x'}, \mathcal{F}_0)$ with x' in some $Z_0(\mathbb{F}_{q^{n'}})$ and $n' < n$. To prove this, one passes through x a curve X_0 such that $(X_0, \mathcal{F}_0|_{X_0})$ is not too complex, and one applies to this curve the results of Section 2. More precisely, one constructs a smooth affine curve X_0, a point $\widetilde{x} \in X_0(\mathbb{F}_{q^n})$, and an $\mathbb{F}_q$-morphism $\varphi : X_0 \to Z_0$ sending $\widetilde{x}$ to x. The construction does not guarantee that the curve X_0 is connected. Replace it by its connected component X_1 such that $\widetilde{x} \in X_1(\mathbb{F}_{q^n})$. The curve X_1 might not be absolutely irreducible over $\mathbb{F}_q$, but there is a bound D, depending only on Z_0, such that X_1 is absolutely irreducible over some $\mathbb{F}_{q^d}$ with $d \leq D$. It is to $X_1/\mathbb{F}_{q^d}$ that the results of Section 2 will be applied. To simplify the exposition of the strategy, we pretend here, falsely, that X_0 is absolutely irreducible over $\mathbb{F}_q$.

Choose a connected etale cover Z' of Z on which $\mathcal{F}$ becomes tamely ramified. This is an enormous waste, but is done only once. Let X' be a connected component of the inverse image of Z' on X. The complexity of $(X_0, \varphi^*\mathcal{F}_0)$ will be measured by $b_1(X')$. We will control it by Katz's theorem [K]. Since only curves are involved here, one could instead use Bombieri [B], on which Katz bases himself, and estimate the number of connected components by the product of the degrees of equations of X', but this would only repeat in a simple case Katz's argument. One obtains a bound for the complexity which is polynomial in n. As soon as n is large enough, the integer N of (0.9.2) satisfies $N < n$, and

$$\mathrm{Tr}(F_x, \mathcal{F}_0) = \mathrm{Tr}(F_{\widetilde{x}}, \varphi^*\mathcal{F}_0)$$

is in the extension of $\mathbb{Q}$ generated by the

$$\mathrm{Tr}(F_{x'}, \varphi^*\mathcal{F}_0) = \mathrm{Tr}(F_{\varphi(x')}, \mathcal{F}_0)$$

for $x' \in X_0(\mathbb{F}_{q^{n'}})$ and $n' \leq N < n$.

I thank H. Esnault for a careful reading of the text and for her suggestions, which I hope have made it more readable.

1. Irreducibility and extensions of the base field

1.1

Let Z_0 be a normal absolutely irreducible scheme of finite type over $\mathbb{F}_q$, let $Z := Z_0 \otimes_{\mathbb{F}_q} \mathbb{F}$, and let x be a geometric point of Z. It defines a geometric point, still denoted x, of Z_0. The fundamental group of Z_0 is an extension

$$1 \to \pi_1(Z, x) \to \pi_1(Z_0, x) \to \mathrm{Gal}(\mathbb{F}/\mathbb{F}_q) \to 1.$$

Taking the fiber product with $W(\mathbb{F}/\mathbb{F}_q) = \mathbb{Z} \subset \mathrm{Gal}(\mathbb{F}/\mathbb{F}_q) = \widehat{\mathbb{Z}}$, one obtains the Weil group $W(Z_0, x)$:

$$1 \to \pi_1(Z, x) \to W(Z_0, x) \to \mathbb{Z} \to 1.$$

The functor "fiber at x" is an equivalence from the category of smooth $\overline{\mathbb{Q}}_\ell$-Weil sheaves on Z_0 to that of continuous linear representations of $W(Z_0, x)$ on a finite-dimensional $\overline{\mathbb{Q}}_\ell$-vector space.

1.2

If V is an irreducible $\overline{\mathbb{Q}}_\ell$-representation of $W(Z_0, x)$, its restriction to $\pi_1(Z, x)$ is a sum of pairwise nonisomorphic irreducible representations of $\pi_1(Z, x)$, permuted transitively by $W(Z_0, x)/\pi_1(Z, x) = \mathbb{Z}$. Let n be the number of summands, and put $Z_1 := Z_0 \otimes_{\mathbb{F}_q} \mathbb{F}_{q^n}$; $W(Z_1, x)$ is the inverse image in $W(Z_0, x)$ of $n\mathbb{Z}$. If S is one of the irreducible summands of the restriction of V to $\pi_1(Z, x)$, then S is a representation of $W(Z_1, x)$ and

$$V = \mathrm{Ind}_{W(Z_1,x)}^{W(Z_0,x)}(S).$$

If $\mathcal{V}$ (respectively $\mathcal{S}$) is the $\overline{\mathbb{Q}}_\ell$-Weil sheaf corresponding to V (respectively S), then $\mathcal{V}$ is the direct image, under $Z_1 \to Z_0$, of $\mathcal{S}$.

1.3

Now let V be a semisimple representation of $W(Z_0, x)$. The quotient $\mathbb{Z}$ of $W(Z_0, x)$ permutes the isomorphism classes of simple constituents of the restriction of V to $\pi_1(Z, x)$. Let A be the set of orbits, and in each orbit a choose a representative S'_a; let $n(a)$ be the number of elements of the orbit. If $Z_a = Z_0 \otimes_{\mathbb{F}_q} \mathbb{F}_{q^{n(a)}}$, the representation S'_a extends to a representation S_a of $W(Z_a, x)$, and S_a is unique up to twist by a character of the quotient $\mathbb{Z}$ of $W(Z_a, x)$. We choose the twist so that the character $\det(S_a)$ of $W(Z_a, x)$ has finite order.

Apply 1.2 to the irreducible constituents of V, and collect the occurrences of S_a. One obtains a decomposition

$$V = \bigoplus_a \mathrm{Ind}_{W(Z_a,x)}^{W(Z_0,x)}(S_a \otimes W_a), \tag{1.3.1}$$

where W_a is a representation of $W(Z_a, x)$ trivial on $\pi_1(Z, x)$. This description of V is unique, except that one may

a) twist S_a by a finite-order character of $\mathbb{Z} = W(Z_a, x)/\pi_1(Z, x)$ and W_a by the inverse character;

b) replace S_a by its conjugate $S_a^{(i)}$ by an element i of $W(Z_0, x)/W(Z_a, x) = \mathbb{Z}/n(a)$.

1.4

Passing from the language of group representations to that of sheaves: S_a (respectively $S_a^{(i)}$) corresponds to $\mathcal{S}_{a,1}$ (respectively $\mathcal{S}_{a,1}^{(i)}$) on Z_a, W_a to $\mathcal{W}_a$ on $\mathrm{Spec}(\mathbb{F}_{q^{n(a)}})$, and, if p_a is the projection from Z_a to Z_0, and pr_a the projection from Z_a to $\mathrm{Spec}(\mathbb{F}_{q^{n(a)}})$, (1.3.1) becomes a decomposition of the $\overline{\mathbb{Q}}_\ell$-Weil sheaf $\mathcal{V}$ corresponding to V:

$$\mathcal{V} = \bigoplus_{a \in A} p_{a*}\left(\mathcal{S}_a \otimes \mathrm{pr}_a^* \mathcal{W}_a\right). \tag{1.4.1}$$

In this decomposition, the $\det(\mathcal{S}_a)$ have finite order, and the inverse images $\mathcal{S}_a^{(i)}$ on Z of the $\mathcal{S}_{a,1}^{(i)}$ ($a \in A$, $i \in \mathbb{Z}/n(a)$) are irreducible pairwise nonisomorphic $\overline{\mathbb{Q}}_\ell$-sheaves.

1.5

Let $\mathcal{F}_0$ be smooth irreducible of finite-order determinant on Z_0 as in 1.1. By 1.2, the inverse image $\mathcal{F}$ of $\mathcal{F}_0$ on $Z := Z_0 \otimes_{\mathbb{F}_q} \mathbb{F}$ is a direct sum of nonisomorphic smooth irreducible sheaves $\mathcal{S}^{(i)}$, permuted transitively by $\mathbb{Z} = W(\mathbb{F}/\mathbb{F}_q)$. Let n be the number of the $\mathcal{S}^{(i)}$. In the remainder of this section, we give a proof of the following statement of Lafforgue [L] VII 7, cf. 0.7.

Theorem 1.6 (Lafforgue). At every closed point x_0 of Z_0, the eigenvalues of F_{x_0} acting on $\mathcal{F}_0$ are Weil numbers of weight 0.

If U_0 is a nonempty open subset of Z_0, the restriction of $\mathcal{F}_0$ to U_0 is still irreducible. Covering Z_0 by affine opens, one reduces to the case where, and we will assume that, Z_0 is affine.

Lemma 1.7. There exists a connected Galois etale cover Z' of Z such that if the inverse image Y' of Z' by a morphism $f : Y \to Z$ is connected, then $\mathcal{F}$ and $f^*\mathcal{F}$ have the same monodromy.

Fix a base point a of Y, and still denote by a its image in Z. There is a morphism

$$\pi_1(Y, a) \longrightarrow \pi_1(Z, a),$$

and the conclusion "same monodromy" means that $\pi_1(Y, a)$ maps onto the image of $\pi_1(Z, a)$ in $\mathrm{GL}(\mathcal{F}_a)$. It implies that the $f^*\mathcal{S}^{(i)}$ remain smooth irreducible and pairwise nonisomorphic.

Proof of 1.7. The monodromy group M, the image of $\pi_1(Z,a)$ in $\mathrm{GL}(\mathcal{F}_a)$, is a closed subgroup of a group $\mathrm{GL}(n,\mathcal{O}_\lambda)$, hence is a compact ℓ-adic Lie group. It therefore admits an open normal pro-ℓ subgroup H, and the intersection H' of the kernels of the homomorphisms $H \to \mathbb{Z}/\ell$ is again an open normal subgroup. Take Z' to be the cover corresponding to H'. Suppose Y' connected. This is equivalent to saying that $\pi_1(Y,a)$ maps onto M/H'. Since every closed subgroup of H which maps onto H/H' coincides with H, $\pi_1(Y,a)$ maps onto M.

1.8

Choose a closed immersion of Z_0 into an affine space E_0 over $\mathbb{F}_q$, and choose Z' as in 1.7. Let G_0 be the $\mathbb{F}_q$-scheme parametrizing affine linear subspaces of codimension $\dim(Z_0)-1$ in E_0. Extend the base field from $\mathbb{F}_q$ to $\mathbb{F}$ and apply to the composite morphism

$$f : Z' \to Z \hookrightarrow E$$

the Bertini theorem in the form given by Jouanolou [J], Th. 6.3(iv), p. 67. One obtains a nonempty open U of the smooth irreducible scheme $G = G_0 \otimes_{\mathbb{F}_q} \mathbb{F}$, such that for every $u \in U(\mathbb{F})$, corresponding to an affine linear subspace L_u of E, the inverse image $f^{-1}(L_u)$ of Z' on $Z \cap L_u$ is irreducible. Note that this irreducibility implies that of $Z \cap L_u$. The set of u such that $Z \cap L_u$ is not a smooth curve has dimension strictly smaller than that of U. Thus, after shrinking U, one may suppose that the $Z \cap L_u$ are smooth curves. After further shrinking U, one may also suppose that U is of the form $U_0 \otimes_{\mathbb{F}_q} \mathbb{F}$, for U_0 an open subset of G_0.

The number of $\mathbb{F}_{q^m}$-points of U_0 is, as $m \to \infty$, of the form

$$q^{\dim(U)m}\left(1 + O(q^{-1/2})\right)$$

(Lang-Weil, or the trace formula plus the Weil conjecture). As soon as m is large enough, $U_0(\mathbb{F}_{q^m})$ is therefore nonempty. This allows one to choose m prime to n such that $U_0(\mathbb{F}_{q^m})$ is nonempty.

For $\mathcal{F}_0$ on Z_0 to satisfy the conclusion of 1.6, it is necessary and sufficient that this conclusion be satisfied by the inverse image of $\mathcal{F}_0$ on $Z_0 \otimes_{\mathbb{F}_q} \mathbb{F}_{q^m}$. Replacing $\mathbb{F}_q$ by $\mathbb{F}_{q^m}$, Z_0 by $Z_0 \otimes_{\mathbb{F}_q} \mathbb{F}_{q^m}$, and $\mathcal{F}_0$ by its inverse image, one is reduced to proving 1.6 under the additional hypothesis that U_0 contains an $\mathbb{F}_q$-point u_0. The plane section $Z_0 \cap L_{u_0}$ of Z_0 is then a smooth absolutely irreducible curve Y_0. Extend the base field to $\mathbb{F}$. Since u_0 is in U_0, 1.7 ensures that the restrictions of the $\mathcal{S}^{(i)}$ to Y are irreducible and pairwise nonisomorphic. Because m is prime to n, they are permuted by $W(\mathbb{F}/\mathbb{F}_q)$: the restriction of $\mathcal{F}_0$ to Y_0 is irreducible. Apply 0.6(i) to it: for every closed point y_0 of $Y_0 \subset Z_0$, the eigenvalues of F_{y_0} acting on $\mathcal{F}_0$ are Weil numbers of weight 0. One concludes by the following proposition.

Proposition 1.9. Let $\mathcal{F}_0$ be a smooth $\overline{\mathbb{Q}}_\ell$-Weil sheaf on a connected scheme X_0 over $\mathbb{F}_q$. The following conditions are equivalent:

(i) there exists a closed point x_0 of X_0 such that the eigenvalues of F_{x_0} acting on $\mathcal{F}_0$ are Weil numbers of weight 0;

(ii) the same assertion holds at all closed points of X_0.

Proof, for X_0 a smooth connected curve. Suppose that $\mathcal{F}_0$ satisfies (i). Every irreducible subquotient $\mathcal{F}'_0$ still satisfies (i), and it suffices to verify (ii) for each $\mathcal{F}'_0$. Let $\mathcal{F}''_0$ be a twist of $\mathcal{F}'_0$ whose determinant has finite order. By 0.6(i), $\mathcal{F}''_0$ satisfies (ii). Testing at x_0, one sees that the twist is by a Weil number of weight 0, and the validity of (ii) is invariant under such a twist.

Proof, general case. For any closed point x'_0 of X_0, there is a sequence of smooth connected curves Y_0^j and morphisms $f^j : Y_0^j \to X_0$ $(1 \leq j \leq N)$ such that x_0 is in the image of Y_0^1, x'_0 is in the image of Y_0^N, and the images of Y_0^j and Y_0^{j+1} intersect nontrivially for $1 \leq j < N$. If (i) holds at x_0, the first part of the proof shows that (ii) holds for the inverse image of $\mathcal{F}_0$ on Y_0^1, hence that (i) holds at every closed point of X_0 in the image of Y_0. Similarly, if (i) holds at the closed points in $f(Y_j)$, and hence at at least one closed point of $f(Y_{j+1})$, then (i) holds at the closed points in $f(Y_{j+1})$, and one eventually reaches x'_0.

2. The case of curves: quantitative version

2.1

Let X be a smooth connected curve over an algebraically closed field of characteristic p, let $\overline{X}$ be the smooth projective completion of X, and let S be the set of points at infinity: $X = \overline{X} - S$. Let $\mathcal{F}$ be a smooth $\overline{\mathbb{Q}}_\ell$-sheaf of rank r on X. Put

$$\chi_c(X, \mathcal{F}) := \sum (-1)^i \dim H_c^i(X, \mathcal{F}).$$

This Euler-Poincare characteristic is given by the Grothendieck-Ogg-Shafarevich formula

$$\chi_c(X, \mathcal{F}) = r\chi_c(X) - \sum_{s \in S} \mathrm{Sw}_s(\mathcal{F}). \tag{2.1.1}$$

If the curve $\overline{X}$ has genus g, then

$$\chi_c(X) = 2 - 2g - |S|. \tag{2.1.2}$$

The term $\mathrm{Sw}_s(\mathcal{F})$ is the Swan conductor of $\mathcal{F}$ at s.

Let K_s be the fraction field of the henselization of X at s. The reader who prefers may replace "henselization" by "completion." This does not change $\mathrm{Gal}(\overline{K}_s/K_s)$, for $\overline{K}_s$ an algebraic closure of K_s. The $\overline{\mathbb{Q}}_\ell$-sheaves on $\mathrm{Spec}(K_s)$ are identified with continuous $\overline{\mathbb{Q}}_\ell$-linear representations V of $\mathrm{Gal}(\overline{K}_s/K_s)$. The conductor $\mathrm{Sw}_s(\mathcal{F})$ is the Swan conductor $\mathrm{Sw}(V)$ of the linear representation V of $\mathrm{Gal}(\overline{K}_s/K_s)$ corresponding to the inverse image of $\mathcal{F}$ on $\mathrm{Spec}(K_s)$.

The conductor $\mathrm{Sw}(V)$ is additive in V, for short exact sequences. Let $\mathrm{Gal}(\overline{K}_s/K_s)^\beta$ denote the ramification subgroups of $\mathrm{Gal}(\overline{K}_s/K_s)$ in upper numbering. Define $\alpha(V)$ as the infimum of the $\beta > 0$ such that $\mathrm{Gal}(\overline{K}_s/K_s)^\beta$ acts trivially on V. If the representation V is irreducible, then

$$\mathrm{Sw}(V) = \dim(V)\,\alpha(V) \qquad (V \text{ irreducible}) \tag{2.1.3}$$

(a consequence of Serre [S] 19.1 and of the definition of the Herbrand functions relating the lower numbering to the upper numbering). If V comes from $\mathcal{F}$ on X, we put $\alpha_s(\mathcal{F}) := \alpha(V)$.

It follows from (2.1.3) by additivity that

$$\mathrm{Sw}(V) \leq \dim(V)\,\alpha(V). \tag{2.1.4}$$

Let P be the pro-p Sylow subgroup of $\mathrm{Gal}(\overline{K}_s/K_s)$. Since V is ℓ-adic, it acts on V through a finite quotient $\overline{P}$. Both $\mathrm{Sw}(V)$ and $\alpha(V)$ vanish if and only if V is tame, i.e. if P acts trivially. Otherwise, $\alpha(V)$ is the largest rational number such that $\overline{P}^{\alpha(V)}$ is nontrivial. The importance for us of the invariant $\alpha(V)$ comes from the observation that

$$\alpha(V \otimes W) \leq \sup(\alpha(V), \alpha(W)). \tag{2.1.5}$$

Poincare duality places $H_c^2(X, \mathcal{F})$ and $H^0(X, \mathcal{F}^\vee(1))$ in perfect duality. In particular, $H_c^2(X, \mathcal{F})$ has rank at most r, and has rank r if $\mathcal{F}$ is constant. If the curve X is affine, then $H_c^0(X, \mathcal{F}) = 0$, and (2.1.1), (2.1.4) give

$$\dim H_c^1(X, \mathcal{F}) \leq r\left(b_1(X) + \sum \alpha_s(\mathcal{F})\right) \qquad (X \text{ affine}), \tag{2.1.6}$$

where we have put

$$b_1(X) = \dim H^1_c(X, \overline{\mathbb{Q}}_\ell) = 2g + |S| - 1 \qquad (X \text{ affine}). \tag{2.1.7}$$

2.2

Let $X_0 = \overline{X}_0 - S_0$ be a curve over $\mathbb{F}_q$ as in 0.0, and let $\mathcal{F}_0$ be a smooth $\overline{\mathbb{Q}}_\ell$-Weil sheaf of rank r on X_0. Assume $\mathcal{F}_0$ algebraic, in the sense of 0.9. We do not assume $\mathcal{F}_0$ irreducible. Let E be a Galois extension of $\mathbb{Q}$ in $\overline{\mathbb{Q}}_\ell$ containing the $\mathrm{Tr}(F_x, \mathcal{F}_0)$ for x in any $X_0(\mathbb{F}_{q^n})$, $n \geq 1$. By 0.6, for every $\sigma \in \mathrm{Gal}(E/\mathbb{Q})$, there is a smooth semisimple $\overline{\mathbb{Q}}_\ell$-Weil sheaf $\sigma(\mathcal{F}_0)$ such that, for every x in $X_0(\mathbb{F}_{q^n})$,

$$\mathrm{Tr}(F_x, \sigma(\mathcal{F}_0)) = \sigma(\mathrm{Tr}(F_x, \mathcal{F}_0)).$$

We write $\sigma(\mathcal{F})$ for the inverse image of $\sigma(\mathcal{F}_0)$ on X. This notation is reasonable because $\sigma(\mathcal{F})$, which is defined up to isomorphism, depends only on $\mathcal{F}$; this follows from (1.4.1) and a compatibility between "σ" and twisting. For each $s \in S$, one has

$$\mathrm{Sw}_s(\sigma\mathcal{F}) = \mathrm{Sw}_s(\mathcal{F}), \qquad \alpha_s(\sigma\mathcal{F}) = \alpha_s(\mathcal{F}). \tag{2.2.1}$$

Sketch of proof. One may suppose $\mathcal{F}_0$ irreducible. If $\mathcal{F}_0$ corresponds to an automorphic representation π, by the rationalized correspondence, then π is defined over E and $\sigma\mathcal{F}_0$ corresponds to $\sigma\pi$. At each $s \in S_0$, $\mathcal{F}_0$ defines a representation of the local Weil group. By the local correspondence, its Frobenius-semisimplification ([De1] 8.6) corresponds to the local component π_s of π. Note that isomorphism classes of continuous $\overline{\mathbb{Q}}_\ell$-linear F-semisimple representations of the local Weil group are in bijection with isomorphism classes of $\overline{\mathbb{Q}}_\ell$-linear F-semisimple representations, loc. cit., of the Weil-Deligne group. The latter are purely algebraic objects. If π_s is defined over E, then the corresponding representation will be also, and conjugation by σ is compatible with the rationalized correspondence. In particular, $\mathcal{F}_0$ and $\sigma(\mathcal{F}_0)$ give rise to representations of the wild inertia group at s having the same kernel, and one is reduced to (2.1.3).

For our main result, the following more elementary lemma suffices.

Lemma 2.3. If $\mathcal{F}$ is tamely ramified, then $\sigma(\mathcal{F})$ is also tamely ramified.

Proof. The image under σ of the L-function of $\mathcal{F}_0$,

$$L(X_0, \mathcal{F}_0, t) = \prod_{x \in |X_0|} \det\left(1 - F_{x_0} t^{\deg(x_0)}, \mathcal{F}_0\right)^{-1},$$

is the L-function of $\sigma(\mathcal{F}_0)$. This L-function is a rational function, and by Grothendieck's cohomological interpretation of L-functions the order of its pole at $t = \infty$ is $-\chi_c(\mathcal{F})$. Thus $\chi_c(\mathcal{F}) = \chi_c(\sigma\mathcal{F})$. Since $\mathcal{F}$ is tamely ramified, the $\mathrm{Sw}_s(\mathcal{F})$ are zero. Comparing (2.1.1) for $\mathcal{F}$ and $\sigma\mathcal{F}$, one sees that the sum of the $\mathrm{Sw}_s(\sigma\mathcal{F})$, hence each of them, is zero, i.e. that $\sigma(\mathcal{F})$ is tamely ramified.

2.4

Suppose X_0 affine, and let $\mathcal{F}_0, \mathcal{G}_0$ be two smooth semisimple Weil sheaves of rank r on X_0. Write $\log_q^+$ for the function $\sup(0, \log_q)$ and $[\ ]$ for the integer-part function. Put

$$\alpha_s(\mathcal{F}, \mathcal{G}) := \alpha_s(\mathcal{F} \oplus \mathcal{G}) = \sup(\alpha_s(\mathcal{F}), \alpha_s(\mathcal{G})), \tag{2.4.1}$$

$$N_0 = 2\log_q^+\left(2r^2\left(b_1(X) + \sum_{s \in S} \alpha_s(\mathcal{F}, \mathcal{G})\right)\right), \tag{2.4.2}$$

$$N = [N_0] + 2r. \tag{2.4.3}$$

Proposition 2.5. If for every integer $n \leq N$ and every $x \in X(\mathbb{F}_{q^n})$ one has

$$\mathrm{Tr}(F_x, \mathcal{F}_0) = \mathrm{Tr}(F_x, \mathcal{G}_0),$$

then $\mathcal{F}_0$ is isomorphic to $\mathcal{G}_0$.

Proof. Apply (1.3), (1.4) to $\mathcal{F}_0 \oplus \mathcal{G}_0$. One obtains decompositions (1.4.1)

$$\mathcal{F}_0 = \bigoplus_{a \in A} p_{a*}\left(\mathcal{S}_{a,1} \otimes \mathrm{pr}_a^* \mathcal{W}_a\right), \tag{2.5.1}$$

$$\mathcal{G}_0 = \bigoplus_{a \in A} p_{a*}\left(\mathcal{S}_{a,1} \otimes \mathrm{pr}_a^* \mathcal{W}'_a\right). \tag{1}$$

In (2.5.1), to each $a \in A$ is attached an integer $n(a) \geq 1$; $\mathcal{S}_{a,1}$ is a smooth $\overline{\mathbb{Q}}_\ell$-Weil sheaf on $X_a := X_0 \otimes_{\mathbb{F}_q} \mathbb{F}_{q^{n(a)}}$; $\mathcal{W}_a$ and $\mathcal{W}'_a$ are $\overline{\mathbb{Q}}_\ell$-Weil sheaves on $\mathrm{Spec}(\mathbb{F}_{q^{n(a)}})$; and p_a and pr_a are the projections from X_a to X_0 and to $\mathrm{Spec}(\mathbb{F}_{q^{n(a)}})$. Let $\mathcal{S}_{a,1}^{(i)}$ denote the image of $\mathcal{S}_{a,1}$ by the i-th power of the Frobenius $F \in \mathrm{Gal}(\mathbb{F}_{q^{n(a)}}/\mathbb{F}_q)$, and omit the subscript 1 to indicate the inverse image from X_a to X. By (1.4), the $\mathcal{S}_{a,1}$, $\mathcal{W}_a$, and $\mathcal{W}'_a$ in (2.5.1) satisfy

(i) $\det(\mathcal{S}_{a,1})$ has finite order;

(ii) the $\mathcal{S}_a^{(i)}$ ($a \in A$, $i \in \mathbb{Z}/n(a)$) are smooth irreducible $\overline{\mathbb{Q}}_\ell$-sheaves on X, pairwise nonisomorphic;

(iii) for each a, $\mathcal{W}_a$ or $\mathcal{W}'_a$ is nonzero.

Since $\mathcal{S}_a^{(i)}$ is a direct factor of $\mathcal{F}$ or $\mathcal{G}$, one has at every point $s \in S$

$$\alpha_s(\mathcal{S}_a^{(i)}) \leq \sup(\alpha_s(\mathcal{F}), \alpha_s(\mathcal{G})).$$

Let $A(n)$ denote the set of a in A such that $n(a) \mid n$.

Lemma 2.6. If $n > N_0$ (2.4.2), the functions on $X_0(\mathbb{F}_{q^n})$

$$t_{a,i} : x \longmapsto \mathrm{Tr}(F_x, \mathcal{S}_{a,1}^{(i)}) \qquad (a \in A(n),\ i \in \mathbb{Z}/n(a))$$

are linearly independent.

Proof. The functions $t_{a,i}$ take values in a number field $E \subset \overline{\mathbb{Q}}_\ell$. Embed E into $\mathbb{C}$ to treat them as complex-valued functions. The idea of the proof is to show that they are nearly orthogonal in $L^2(X_0(\mathbb{F}_{q^n}))$.

By 0.6(i), $\mathcal{S}_{a,1}^{(i)}$ is pure of weight 0. The complex conjugate function of $t_{a,i}$ is therefore

$$x \longmapsto \mathrm{Tr}(F_x, (\mathcal{S}_{a,1}^{(i)})^\vee),$$

and the scalar product $\langle t_{b,j}, t_{a,i}\rangle = \sum \overline{t_{b,j}(x)} t_{a,i}(x)$ is

$$\langle t_{b,j}, t_{a,i}\rangle = \sum \mathrm{Tr}(F_x, \mathcal{H}om(\mathcal{S}_{a,1}^{(i)}, \mathcal{S}_{b,1}^{(j)})).$$

By the trace formula, this sum is the trace of the Frobenius $F \in W(\mathbb{F}/\mathbb{F}_{q^n})$:

$$\langle t_{b,j}, t_{a,i}\rangle = \sum (-1)^k \,\mathrm{Tr}\left(F, H_c^k\left(X, \mathcal{H}om(\mathcal{S}_a^{(i)}, \mathcal{S}_b^{(j)})\right)\right).$$

Since X is affine, H_c^0 is zero. The "dominant" term is given by H_c^2: it is q^n if $(a,i) = (b,j)$ and zero otherwise. The promised almost orthogonality comes from the fact that the term $k = 1$ is

$O(q^{n/2})$ for large n. More precisely, since $\mathcal{H}om(\mathcal{S}_{a,1}^{(i)}, \mathcal{S}_{b,1}^{(j)})$ is pure of weight 0, the eigenvalues of F on its H_c^1 have absolute value $q^{n/2}$ or 1, and the term $k=1$ is bounded in absolute value by $q^{n/2}$ times

$$\dim H_c^1 \leq \dim \mathcal{S}_b^{(j)} \cdot \dim \mathcal{S}_a^{(i)} \cdot \left(b_1(X) + \sum \alpha_s(\mathcal{F}, \mathcal{G})\right).$$

Suppose that there is a linear dependence relation $\sum \lambda_{b,j} t_{b,j} = 0$. Choose a, i such that $|\lambda_{a,i}|$ is maximal among the $|\lambda_{b,j}|$. Dividing by $\lambda_{a,i}$, one may suppose $\lambda_{a,i} = 1$ and $|\lambda_{b,j}| \leq 1$ for $b \in A(n)$ and $j \in \mathbb{Z}/n(b)$. Then

$$0 = \left\langle \sum \lambda_{b,j} t_{b,j}, t_{a,i} \right\rangle = \sum_{b,j} \lambda_{b,j} \langle t_{b,j}, t_{a,i} \rangle = q^n + \text{remainder},$$

where the absolute value of the remainder is bounded by

$$\sum_{b,j} |\lambda_{b,j}| \left| \operatorname{Tr}\left(F, H_c^1(X, \mathcal{H}om(\mathcal{S}_a^{(i)}, \mathcal{S}_b^{(j)}))\right) \right| \leq q^{n/2}\, 2r^2 \left(b_1(X) + \sum \alpha_s(\mathcal{F}, \mathcal{G})\right).$$

The hypothesis on n ensures $|\text{remainder}| < q^n$: contradiction.

Corollary 2.7. Let F denote the Frobenius of $W(\mathbb{F}/\mathbb{F}_{q^n})$. If $n > N_0$ and $\operatorname{Tr}(F_x, \mathcal{F}_0) = \operatorname{Tr}(F_x, \mathcal{G}_0)$ for $x \in X_0(\mathbb{F}_{q^n})$, then

$$\operatorname{Tr}(F, \mathcal{W}_a) = \operatorname{Tr}(F, \mathcal{W}'_a)$$

for each $a \in A(n)$.

Proof. The decompositions (2.5.1) give the identity between trace functions

$$\sum t_{a,i} \operatorname{Tr}(F, \mathcal{W}_a) = \sum t_{a,i} \operatorname{Tr}(F, \mathcal{W}'_a),$$

and one applies 2.6.

2.8. End of the proof of 2.5

It remains to show that, for each a, and for F the geometric Frobenius generator of $W(\mathbb{F}/\mathbb{F}_{q^{n(a)}})$, F has the same multiset of eigenvalues on $\mathcal{W}_a$ and $\mathcal{W}'_a$. By 2.7, if n is divisible by $n(a)$ and

$$[N_0] + 1 \leq n \leq N = [N_0] + 2r,$$

then

$$\operatorname{Tr}(F^{n/n(a)}, \mathcal{W}_a) = \operatorname{Tr}(F^{n/n(a)}, \mathcal{W}'_a).$$

There are at least $[2r/a]$ such values of n; $\mathcal{W}_a$ and $\mathcal{W}'_a$ are of dimension at most $[r/a]$; it remains to apply the following lemma.

Lemma 2.9. Let α_i be k distinct nonzero numbers. If the λ_i satisfy, for a suitable m,

$$\sum \lambda_i \alpha_i^r = 0 \qquad \text{for } m \leq r < m+k, \tag{2.9.1}$$

then all the λ_i are zero.

Proof. Equation (2.9.1) can be rewritten

$$\sum (\lambda_i \alpha_i^m) \alpha_i^s = 0 \qquad \text{for } 0 \leq s < k,$$

and the Vandermonde determinant $\det(\alpha_i^s)$ is nonzero.

In the proposition which follows, X_0 is assumed affine, and $\mathcal{F}_0$ is a smooth Weil sheaf of rank r on X_0. Assume $\mathcal{F}_0$ algebraic in the sense of 0.9. Put

$$N_0 = 2 \log_q^+ \left(2r^2 \left(b_1(X) + \sum \alpha_s(\mathcal{F})\right)\right), \qquad N = [N_0] + 2r.$$

Proposition 2.10. Let E_0 be the extension of $\mathbb{Q}$ in $\overline{\mathbb{Q}}_\ell$ generated by the $\mathrm{Tr}(F_x, \mathcal{F}_0)$ for x in $X_0(\mathbb{F}_{q^n})$ with $n \leq N$. Then, for every n and every $x \in X_0(\mathbb{F}_{q^n})$, $\mathrm{Tr}(F_x, \mathcal{F}_0)$ is in E_0.

Proof. After semisimplifying $\mathcal{F}_0$, one reduces to the case where $\mathcal{F}_0$ is semisimple. Let E be as in 2.2, a sufficiently large Galois extension of $\mathbb{Q}$ in $\overline{\mathbb{Q}}_\ell$. We must show that for $\sigma \in \mathrm{Gal}(E/E_0)$, the $\mathrm{Tr}(F_x, \mathcal{F}_0)$ are fixed by σ. This is equivalent to

$$\mathrm{Tr}(F_x, \mathcal{F}_0) = \mathrm{Tr}(F_x, \sigma(\mathcal{F}_0)). \tag{2.10.1}$$

By hypothesis, (2.10.1) is true if x is in some $X(\mathbb{F}_{q^n})$ with $n \leq N$. By 2.5 and by the fact that $\alpha_s(\mathcal{F}_0) = \alpha_s(\sigma(\mathcal{F}_0))$, $\mathcal{F}_0$ is isomorphic to $\sigma(\mathcal{F}_0)$. The assertion follows.

2.11. Variants

Keep the hypotheses and notation of 2.5. Let $q : X' \to X$ be a connected etale cover of X on which the inverse images of $\mathcal{F}$ and $\mathcal{G}$ are tamely ramified. For every $\overline{\mathbb{Q}}_\ell$-sheaf $\mathcal{H}$ on X whose inverse image on X' is tamely ramified, the morphism

$$q : H^*(X, \mathcal{H}) \to H^*(X', q^*\mathcal{H})$$

is injective, because its composite with Tr_q is multiplication by the degree of the cover. Thus

$$\dim H^1_c(X, \mathcal{H}) \leq \dim H^1_c(X', \mathcal{H}) \leq \mathrm{rank}(\mathcal{H}) b_1(X').$$

If one repeats the arguments proving 2.5 using this estimate rather than (2.1.6), one obtains the following.

Variant 2.12. Put $N'_0 := 2 \log_q^+(2r^2 b_1(X'))$ and $N' := [N'_0] + 2r$. If for every integer $n \leq N'$ and every $x \in X_0(\mathbb{F}_{q^n})$ one has

$$\mathrm{Tr}(F_x, \mathcal{F}_0) = \mathrm{Tr}(F_x, \mathcal{G}_0),$$

then $\mathcal{F}_0$ is isomorphic to $\mathcal{G}_0$.

Similarly, if $\mathcal{F}_0$ is as in 2.10 and its inverse image by $q : X' \to X$ is tamely ramified, then $\sigma(\mathcal{F}_0)$ has the same property: after a finite extension of the base field $\mathbb{F}_q$, one may suppose that $q : X' \to X$ comes from $q_0 : X'_0 \to X_0$ and apply 2.3. Repeating similarly the arguments which prove 2.10, one obtains the following.

Variant 2.13. Let E_0 be the extension of $\mathbb{Q}$ in $\overline{\mathbb{Q}}_\ell$ generated by the $\mathrm{Tr}(F_x, \mathcal{F}_0)$ for x in $X_0(\mathbb{F}_{q^n})$ with $n \leq N'$, where N' is as in 2.12. Then, for every n and every $x \in X_0(\mathbb{F}_{q^n})$, $\mathrm{Tr}(F_x, \mathcal{F}_0)$ is in E_0.

Remark 2.14. The hypothesis that $\mathcal{F}_0$ is algebraic ensures that E_0 is a finite extension of $\mathbb{Q}$. It is not required in order to prove 2.13. If $E_0 \subset E$ are extensions of $\mathbb{Q}$ in $\overline{\mathbb{Q}}_\ell$, and if P is the set of $\sigma : E \to \overline{\mathbb{Q}}_\ell$ extending the inclusion of E_0 in $\overline{\mathbb{Q}}_\ell$, then

$$E_0 = \{e \in E \mid \sigma(e) = e \text{ for every } \sigma \in \Sigma\},$$

and this observation can replace the Galois argument.

3. Main theorem

Theorem 3.1. Let Z_0 be a scheme of finite type over $\mathbb{F}_q$. If $\mathcal{F}_0$ is an algebraic $\overline{\mathbb{Q}}_\ell$-Weil sheaf (0.9) on Z_0, there exists a finite extension $E \subset \overline{\mathbb{Q}}_\ell$ of $\mathbb{Q}$ such that for every n and every $x \in Z_0(\mathbb{F}_{q^n})$, $\mathrm{Tr}(F_x, \mathcal{F}_0)$ lies in E.

The fact that all these traces lie in E is equivalent to saying that, at every closed point x_0 of Z_0, the local factor of the L-function of $\mathcal{F}_0$ has coefficients in E; see 0.5.

Lemma 3.2. Theorem 3.1 is implied by the special case in which one assumes that

(i) Z_0 is affine, smooth, irreducible, and is endowed with an etale morphism $\varphi : Z_0 \to \mathbb{E}_0^k$ to an affine space over $\mathbb{F}_q$;

(ii) $\mathcal{F}_0$ is smooth, irreducible, and $\det(\mathcal{F}_0)$ has finite order.

Proof. The scheme Z_0 of 3.1 admits a partition into irreducible locally closed parts F_i satisfying (i) and such that $\mathcal{F}_0|_{F_i}$ is smooth. It suffices to treat separately each $(F_i, \mathcal{F}_0|_{F_i})$. One can then treat separately each irreducible subquotient of $\mathcal{F}_0|_{F_i}$, and twist it so that it satisfies (ii).

Lemma 3.3. There exists a connected etale cover $q : Z' \to Z$ such that the monodromy group of $q^*\mathcal{F}$ is a pro-ℓ group.

What matters for us is that the restriction of $q^*\mathcal{F}$ to a curve drawn on Z' is tamely ramified at infinity.

Proof. There exists a finite extension E_λ of $\mathbb{Q}_\ell$ in $\overline{\mathbb{Q}}_\ell$, with ring of integers $\mathcal{O}_\lambda$, and a torsion-free smooth $\mathcal{O}_\lambda$-sheaf $\widetilde{\mathcal{F}}$ such that

$$\mathcal{F} \simeq \widetilde{\mathcal{F}} \otimes_{\mathcal{O}_\lambda} \overline{\mathbb{Q}}_\ell.$$

The reduction of $\widetilde{\mathcal{F}}$ modulo the maximal ideal $\mathfrak{m}_\lambda$ of $\mathcal{O}_\lambda$ is a locally constant etale sheaf, and it suffices to take for Z' a cover of Z on which it becomes constant. If $\mathcal{F}$ has rank r, one can take for Z' a connected component of the cover

$$\operatorname{Isom}\left(\widetilde{\mathcal{F}}/\mathfrak{m}_\lambda\widetilde{\mathcal{F}}, (\mathcal{O}_\lambda/\mathfrak{m}_\lambda)^r\right).$$

On the cover Z', the monodromy of each $\widetilde{\mathcal{F}}/\mathfrak{m}_\lambda^n\widetilde{\mathcal{F}}$ is an ℓ-group.

Proposition 3.4. If the integer n is sufficiently large, then for every $x \in Z_0(\mathbb{F}_{q^n})$ the trace $\operatorname{Tr}(F_x, \mathcal{F}_0)$ is contained in the extension of $\mathbb{Q}$ in $\overline{\mathbb{Q}}_\ell$ generated by the traces $\operatorname{Tr}(F_y, \mathcal{F}_0)$ with $y \in Z_0(\mathbb{F}_{q^m})$ and $m < n$.

If N is such that the integers $n > N$ are "sufficiently large," it follows by induction that every trace $\operatorname{Tr}(F_x, \mathcal{F}_0)$ is contained in the extension of $\mathbb{Q}$ generated by the $\operatorname{Tr}(F_y, \mathcal{F}_0)$ with y in some $Z_0(\mathbb{F}_{q^m})$ with $m \le N$. This extension is finite, and therefore 3.4 implies 3.1.

Let $x \in Z_0(\mathbb{F}_{q^n})$, and try to prove the conclusion of 3.4 for x. We will succeed for n large enough.

Choose a generator of $\mathbb{F}_{q^n}$ over $\mathbb{F}_q$. This choice defines an $\mathbb{F}_{q^n}$-point x' of the affine line $\mathbb{E}_0^1$ over $\mathbb{F}_q$, whose $\operatorname{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$-conjugates are all distinct.

Lemma 3.5. If y is an $\mathbb{F}_{q^n}$-point of $\mathbb{E}_0^1$, there exists an $\mathbb{F}_q$-morphism $P : \mathbb{E}_0^1 \to \mathbb{E}_0^1$, i.e. a polynomial $P \in \mathbb{F}_q[T]$, which sends x' to y and has degree $\le n-1$.

Proof. The $\sigma x'$, for $\sigma \in \operatorname{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$, are n distinct points of the affine line. There is therefore a unique polynomial $P \in \mathbb{F}_{q^n}[T]$, of degree $\le n-1$, which for each σ takes at $\sigma x'$ the value σy. Its uniqueness ensures its Galois invariance, hence that it lies in $\mathbb{F}_q[T]$.

Corollary 3.6. There exists an $\mathbb{F}_q$-morphism $P : \mathbb{E}_0^1 \to \mathbb{E}_0^k$, whose coordinates are polynomials of degree $\le n-1$, which sends $x' \in \mathbb{E}_0^1(\mathbb{F}_{q^n})$ to $\varphi(x) \in \mathbb{E}_0^k(\mathbb{F}_{q^n})$.

This is checked by applying 3.5 to each coordinate of $\varphi(x)$.

3.7

Fix P as in 3.6. Let X''_0 be the fiber product of Z_0 and $\mathbb{E}^1_0$ over $\mathbb{E}^k_0$. Since $\varphi(x) = P(x')$, there exists $\bar{x} \in X''_0(\mathbb{F}_{q^n})$ mapping to x and x'. Let X_0 denote the connected component of X''_0 containing $\bar{x}$:

$$\begin{tikzcd} X_0 \arrow[r, hook] \arrow[dr, "\widetilde{P}"'] & X''_0 \arrow[r, "\varphi''"] \arrow[d] & \mathbb{E}^1_0 \arrow[d, "P"] & \bar{x} \in X_0(\mathbb{F}_{q^n}) \arrow[d] \\ & Z_0 \arrow[r, "\varphi"] & \mathbb{E}^k_0 & x \in Z_0(\mathbb{F}_{q^n}). \end{tikzcd} \tag{3.7.1}$$

The morphism φ'' is etale, of degree over the generic point of $\mathbb{E}^1_0$ at most equal to the degree D of φ. Let k_0 be the field of constants of the curve X_0 over $\mathbb{F}_q$. The degree d of k_0 over $\mathbb{F}_q$ is $\leq D$, because X_0 is of degree $\leq D$ over $\mathbb{E}^1_0$, and d divides n, because X_0 has a point over $\mathbb{F}_{q^n}$. Extend the base field from $\mathbb{F}_q$ to $\mathbb{F}_{q^d}$, and let X_1 (respectively Z_1) be the connected component of $X_0 \otimes_{\mathbb{F}_q} \mathbb{F}_{q^d}$ (respectively of $Z_0 \otimes_{\mathbb{F}_q} \mathbb{F}_{q^d}$) which contains $\bar{x}$ (respectively x):

$$\begin{tikzcd} \bar{x} : \mathrm{Spec}(\mathbb{F}_{q^n}) \arrow[r] \arrow[d, equal] & X_1 \arrow[r, "\sim"] \arrow[d] & X_0 \arrow[d, "\widetilde{P}"] \\ x : \mathrm{Spec}(\mathbb{F}_{q^n}) \arrow[r] & Z_1 \arrow[r] \arrow[d] & Z_0 \arrow[d] \\ & \mathrm{Spec}(\mathbb{F}_{q^d}) \arrow[r] & \mathrm{Spec}(\mathbb{F}_q). \end{tikzcd} \tag{3.7.2}$$

The curve X_1 over $\mathbb{F}_{q^d}$ is smooth and absolutely irreducible. Finally extend the base field from $\mathbb{F}_{q^d}$ to $\mathbb{F}$, to obtain X and Z, and let X' be a connected component of the inverse image X'' of the etale cover Z' of Z:

$$\begin{tikzcd} X' \arrow[r, hook] \arrow[d] & X'' \arrow[r] \arrow[d] & X \\ Z' \arrow[r] & Z. \end{tikzcd} \tag{3.7.3}$$

3.8. Proof of 3.4

The inverse image of $\mathcal{F}$ on X' is tamely ramified. We propose to apply 2.13 to the curve X_1 over $\mathbb{F}_{q^d}$, to $X' \to X$, and to the inverse image $\mathcal{F}_1$ of $\mathcal{F}_0$ on X_1. Let us estimate $b_1(X')$.

The composite $\mathbb{F}$-morphism

$$Z' \to Z \xrightarrow{\varphi} \mathbb{E}^k$$

is affine. It therefore factors through a closed immersion $Z' \hookrightarrow \mathbb{E}^{k+k'}$, and there is a family of A' equations of degree $\leq B$ defining Z' in $\mathbb{E}^{k+k'}$.

The graph Γ_P of $P : \mathbb{E}^1 \to \mathbb{E}^k$ is defined in $\mathbb{E}^{1+k}$ by k equations of degree $\leq n-1$. The fiber product, calculated over $\mathbb{F}$, of $Z' \to \mathbb{E}^k$ and $P : \mathbb{E}^1 \to \mathbb{E}^k$ is the intersection, in $\mathbb{E}^{1+k+k'}$, of the inverse images of $Z' \subset \mathbb{E}^{k+k'}$ and of $\Gamma_P \subset \mathbb{E}^{1+k}$. It is therefore defined, in $\mathbb{E}^{1+k+k'}$, by $A := A' + k$ equations of degree $\leq \sup(B, n-1)$. By Katz [K], this implies an upper bound for the sum of the Betti numbers of this fiber product, and a fortiori of $b_1(X')$, since X' is a connected component of the fiber product.

For $n > B$, one has A equations of degree $\leq n-1$ in an affine space of dimension $k + k' + 1$, and Katz gives

$$b_1(X') \leq 6 \cdot 2^A (An+3)^{k+k'+1}. \tag{3.8.1}$$

Put, as in 2.12,

$$N'_0 = 2\log^+_{q^d}(2r^2 b_1(X')) \qquad \text{and} \qquad N' = [N'_0] + 2r.$$

The bound (3.8.1) is polynomial in n, and d is bounded by D. As soon as n is large enough, one therefore has

$$\frac{n}{d} > N'.$$

Assume n large enough for this inequality to hold.

By (3.7.2), one has $\mathrm{Tr}(F_{\bar{x}}, \mathcal{F}_1) = \mathrm{Tr}(F_x, \mathcal{F}_0)$. By 2.13, this trace is contained in the field generated by the $\mathrm{Tr}(F_y, \mathcal{F}_1)$ for y in $X_1(\mathbb{F}_{q^m})$ with $d \mid m$ and with m/d, the degree of $\mathbb{F}_{q^m}$ over $\mathbb{F}_{q^d}$, at most N'. Again by (3.7.2), these traces are also the $\mathrm{Tr}(F_y, \widetilde{P}^*\mathcal{F}_0)$ for y in $X_0(\mathbb{F}_{q^m})$ with $m/d \leq N'$. We have

$$\mathrm{Tr}(F_y, \widetilde{P}^*\mathcal{F}_0) = \mathrm{Tr}(F_{\widetilde{P}(y)}, \mathcal{F}_0),$$

and $\mathrm{Tr}(F_x, \mathcal{F}_0)$ is in the field generated by the $\mathrm{Tr}(F_y, \mathcal{F}_0)$ with $y \in Z_0(\mathbb{F}_{q^m})$ and

$$\frac{m}{d} \leq N' < \frac{n}{d},$$

and hence $m < n$. This proves 3.4.

Remark 3.9. If one does not assume $\mathcal{F}_0$ algebraic, it remains true that the extension of $\mathbb{Q}$ in $\overline{\mathbb{Q}}_\ell$ generated by the $\mathrm{Tr}(F_x, \mathcal{F}_0)$ for x in any $Z_0(\mathbb{F}_{q^n})$ is an extension of finite type. One may deduce this by twisting from the algebraic case, or repeat the preceding arguments while applying 2.14.

Remark 3.10. Except in the proof of 3.3, our arguments have used the sheaf $\mathcal{F}_0$ only through its inverse images on smooth curves. Theorem 3.1 therefore remains valid if, instead of $\mathcal{F}_0$, one is given "trace" functions t_n on the $X_0(\mathbb{F}_{q^n})$, with values in $\overline{\mathbb{Q}}_\ell$, having the following two properties:

a. For every smooth curve X_0 and every morphism $\varphi : X_0 \to Z_0$, there is a smooth $\overline{\mathbb{Q}}_\ell$-sheaf $\mathcal{F}_0[\varphi]$ on X_0 such that, for every n and every $x \in X_0(\mathbb{F}_{q^n})$,

$$\mathrm{Tr}(F_x, \mathcal{F}_0[\varphi]) = t_n(\varphi(x)).$$

b. There exists a connected etale cover $q : Z' \to Z$ such that for every $\varphi : X_0 \to Z_0$ as in a., if X' denotes the etale cover of X obtained as the inverse image of Z', then the inverse image of $\mathcal{F}_0[\varphi]$ on X' is tamely ramified at infinity.

Hypothesis b. replaces 3.3. The conclusion is that the values of the t_n generate a finite-type extension of $\mathbb{Q}$; a finite extension if the values of the t_n are algebraic.

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